

IMPORTING LIBRARIES

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: pd.read_csv('batsman_runs_ipl.csv')
```

```
Out[2]:
```

	batter	batsman_run
0	A Ashish Reddy	280
1	A Badoni	161
2	A Chandila	4
3	A Chopra	53
4	A Choudhary	25
...	...	...
600	Yash Dayal	0
601	Yashpal Singh	47
602	Younis Khan	3
603	Yuvraj Singh	2754
604	Z Khan	117

605 rows × 2 columns

```
In [3]: pd.read_csv('diabetes.csv')
```

Out[3]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesP
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
...	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

768 rows × 9 columns

```
In [12]: ipl = pd.read_csv('ipl-matches.csv')
```

```
In [11]: movies= pd.read_csv('movies.csv')
```

CREATING DATAFRAME

```
In [7]: #using list

student_data = [
    [100,90,7],
    [78,90,8],
    [90,80,9],
    [75,69,5]
]
pd.DataFrame(student_data,columns=['IQ','MARKS','CGPA'])
```

Out[7]:

	IQ	MARKS	CGPA
0	100	90	7
1	78	90	8
2	90	80	9
3	75	69	5

```
In [10]: #using dictionary
student_dict = {
    "name":["marina","manisha","eshu","sangeetha"],
    "iq":[100,90,60,70],
    "marks" :[90,80,70,67],
    "package":[7,7,8,9]
}
```

```
students = pd.DataFrame(student_dict)
students.set_index("name", inplace = True)
students
```

Out[10]:

	iq	marks	package
name			
marina	100	90	7
manisha	90	80	7
eshu	60	70	8
sangeetha	70	67	9

name			
marina	100	90	7
manisha	90	80	7
eshu	60	70	8
sangeetha	70	67	9

DATAFRAME ATTRIBUTES AN METHODS

In [14]: `# shape`
`movies.shape`

Out[14]: (1629, 18)

In [15]: `#size`
`movies.size`

Out[15]: 29322

In [16]: `#index`
`ipl.index`

Out[16]: RangeIndex(start=0, stop=950, step=1)

In [17]: `#dtypes`
`ipl.dtypes`

Out[17]:

ID	int64
City	object
Date	object
Season	object
MatchNumber	object
Team1	object
Team2	object
Venue	object
TossWinner	object
TossDecision	object
SuperOver	object
WinningTeam	object
WonBy	object
Margin	float64
method	object
Player_of_Match	object
Team1Players	object
Team2Players	object
Umpire1	object
Umpire2	object
dtype:	object

```
In [18]: ipl.columns
```

```
Out[18]: Index(['ID', 'City', 'Date', 'Season', 'MatchNumber', 'Team1', 'Team2',  
              'Venue', 'TossWinner', 'TossDecision', 'SuperOver', 'WinningTeam',  
              'WonBy', 'Margin', 'method', 'Player_of_Match', 'Team1Players',  
              'Team2Players', 'Umpire1', 'Umpire2'],  
              dtype='object')
```

```
In [19]: ipl.values
```

```
Out[19]: array([[1312200, 'Ahmedabad', '2022-05-29', ...,  
                "['WP Saha', 'Shubman Gill', 'MS Wade', 'HH Pandya', 'DA Miller', 'R Tewatia', 'Rashid Khan', 'R Sai Kishore', 'LH Ferguson', 'Yash Dnyal', 'Mohammed Shami']",  
                'CB Gaffaney', 'Nitin Menon'],  
               [1312199, 'Ahmedabad', '2022-05-27', ...,  
                "['YBK Jaiswal', 'JC Buttler', 'SV Samson', 'D Padikkal', 'SO Hetmyer', 'R Parag', 'R Ashwin', 'TA Boult', 'YS Chahal', 'M Prasidh Krishna', 'OC McCoy']",  
                'CB Gaffaney', 'Nitin Menon'],  
               [1312198, 'Kolkata', '2022-05-25', ...,  
                "['Q de Kock', 'KL Rahul', 'M Vohra', 'DJ Hooda', 'MP Stoinis', 'E Lewis', 'KH Pandya', 'PVD Chameera', 'Mohsin Khan', 'Avesh Khan', 'Ravi Bishnoi']",  
                'J Madanagopal', 'MA Gough'],  
               ...,  
               [335984, 'Delhi', '2008-04-19', ...,  
                "['T Kohli', 'YK Pathan', 'SR Watson', 'M Kaif', 'DS Lehmann', 'RA Jadeja', 'M Rawat', 'D Salunkhe', 'SK Warne', 'SK Trivedi', 'MM Patel']",  
                'Aleem Dar', 'GA Pratapkumar'],  
               [335983, 'Chandigarh', '2008-04-19', ...,  
                "['PA Patel', 'ML Hayden', 'MEK Hussey', 'MS Dhoni', 'SK Raina', 'JDP Oram', 'S Badrinath', 'Joginder Sharma', 'P Amarnath', 'MS Gony', 'M Muralitharan']",  
                'MR Benson', 'SL Shastri'],  
               [335982, 'Bangalore', '2008-04-18', ...,  
                "['SC Ganguly', 'BB McCullum', 'RT Ponting', 'DJ Hussey', 'Mohammad Hafeez', 'LR Shukla', 'WP Saha', 'AB Agarkar', 'AB Dinda', 'M Kartik', 'I Sharma']",  
                'Asad Rauf', 'RE Koertzen']], dtype=object)
```

```
In [21]: #head  
movies.head(3)
```

```
Out[21]:
```

	title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum... https://en.w

1	Battalion 609	tt9472208	NaN https://
---	---------------	-----------	--------------

2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum... https://en.w
---	--------------------------------------	-----------	--

```
In [22]: ipl.tail(3)
```

```
Out[22]:
```

	ID	City	Date	Season	MatchNumber	Team1	Team2
947	335984	Delhi	2008-04-19	2007/08	3	Delhi Daredevils	Rajasthan Royals
948	335983	Chandigarh	2008-04-19	2007/08	2	Kings XI Punjab	Chennai Super Kings
949	335982	Bangalore	2008-04-18	2007/08	1	Royal Challengers Bangalore	Kolkata Knight Riders

```
In [23]: #sample
movies.sample()
```

```
Out[23]:
```

	title_x	imdb_id	poster_path
666	Tere Naal Love Ho Gaya	tt2130242	https://upload.wikimedia.org/wikipedia/en/thum... https://en.wik

```
In [26]: #info
movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1629 entries, 0 to 1628
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x               1629 non-null   object
1   imdb_id               1629 non-null   object
2   poster_path           1526 non-null   object
3   wiki_link             1629 non-null   object
4   title_y               1629 non-null   object
5   original_title        1629 non-null   object
6   is_adult              1629 non-null   int64
7   year_of_release       1629 non-null   int64
8   runtime               1629 non-null   object
9   genres                1629 non-null   object
10  imdb_rating           1629 non-null   float64
11  imdb_votes            1629 non-null   int64
12  story                 1609 non-null   object
13  summary               1629 non-null   object
14  tagline               557 non-null    object
15  actors                1624 non-null   object
16  wins_nominations      707 non-null    object
17  release_date          1522 non-null   object
dtypes: float64(1), int64(3), object(14)
memory usage: 229.2+ KB
```

```
In [27]: ipl.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    950 non-null    int64
1   City                  899 non-null    object
2   Date                  950 non-null    object
3   Season                950 non-null    object
4   MatchNumber           950 non-null    object
5   Team1                 950 non-null    object
6   Team2                 950 non-null    object
7   Venue                 950 non-null    object
8   TossWinner            950 non-null    object
9   TossDecision          950 non-null    object
10  SuperOver             946 non-null    object
11  WinningTeam           946 non-null    object
12  WonBy                 950 non-null    object
13  Margin                932 non-null    float64
14  method                19 non-null     object
15  Player_of_Match       946 non-null    object
16  Team1Players           950 non-null    object
17  Team2Players           950 non-null    object
18  Umpire1               950 non-null    object
19  Umpire2               950 non-null    object
dtypes: float64(1), int64(1), object(18)
memory usage: 148.6+ KB
```

```
In [28]: #describe
```

```
movies.describe()
```

```
Out[28]:
```

	is_adult	year_of_release	imdb_rating	imdb_votes
count	1629.0	1629.000000	1629.000000	1629.000000
mean	0.0	2010.263966	5.557459	5384.263352
std	0.0	5.381542	1.567609	14552.103231
min	0.0	2001.000000	0.000000	0.000000
25%	0.0	2005.000000	4.400000	233.000000
50%	0.0	2011.000000	5.600000	1000.000000
75%	0.0	2015.000000	6.800000	4287.000000
max	0.0	2019.000000	9.400000	310481.000000

```
In [30]: #isnull
ipl.isnull().sum()
```

```
Out[30]:
```

ID	0
City	51
Date	0
Season	0
MatchNumber	0
Team1	0
Team2	0
Venue	0
TossWinner	0
TossDecision	0
SuperOver	4
WinningTeam	4
WonBy	0
Margin	18
method	931
Player_of_Match	4
Team1Players	0
Team2Players	0
Umpire1	0
Umpire2	0
dtype:	int64

```
In [31]: ##
movies.isnull().sum()
```

```
Out[31]: title_x          0
         imdb_id         0
         poster_path    103
         wiki_link       0
         title_y         0
         original_title   0
         is_adult         0
         year_of_release  0
         runtime          0
         genres           0
         imdb_rating      0
         imdb_votes       0
         story            20
         summary          0
         tagline          0
         actors           5
         wins_nominations 922
         release_date     107
         dtype: int64
```

```
In [33]: #duplicated ## repeated values
         movies.duplicated().sum()
```

```
Out[33]: np.int64(0)
```

```
In [36]: ipl.duplicated().sum()
```

```
Out[36]: np.int64(0)
```

```
In [41]: #rename function is used to create different values as per preferred rows
         students.rename(columns = {"marks":"percentage","lpa":"CGPA"},inplace = T
```

```
In [42]: students
```

```
Out[42]:
```

	iq	percentage	package
name			
marina	100	90	7
manisha	90	80	7
eshu	60	70	8
sangeetha	70	67	9

MATH METHOD

```
In [43]: students
```

Out[43]:

	iq	percentage	package
name			
marina	100	90	7
manisha	90	80	7
eshu	60	70	8
sangeetha	70	67	9

name			
marina	100	90	7
manisha	90	80	7
eshu	60	70	8
sangeetha	70	67	9

```
In [47]: #sum
students.sum(axis = 1) #1 is for row
```

Out[47]:

name	
marina	197
manisha	177
eshu	138
sangeetha	146

dtype: int64

```
In [48]: students.sum(axis = 0) #0 is for column
```

Out[48]:

iq	320
percentage	307
package	31

dtype: int64

```
In [49]: students.mean(axis=0)
```

Out[49]:

iq	80.00
percentage	76.75
package	7.75

dtype: float64

```
In [50]: students.median(axis = 0)
```

Out[50]:

iq	80.0
percentage	75.0
package	7.5

dtype: float64

```
In [51]: students.std()
```

Out[51]:

iq	18.257419
percentage	10.436315
package	0.957427

dtype: float64

```
In [52]: students.var()
```

Out[52]:

iq	333.333333
percentage	108.916667
package	0.916667

dtype: float64

```
In [54]: students.min(axis = 1)
```

```
Out[54]: name
         marina      7
         manisha     7
         eshu        8
         sangeetha   9
         dtype: int64
```

```
In [55]: students.max()
```

```
Out[55]: iq          100
         percentage   90
         package      9
         dtype: int64
```

```
In [56]: students.duplicated()
```

```
Out[56]: name
         marina      False
         manisha     False
         eshu        False
         sangeetha   False
         dtype: bool
```

```
In [ ]:
```